TAO BodyPose3DNet 3D human skeleton detection

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> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case objective and processing flow

First the people are detected and tracked, then 34 2.5D/3D body key points are estimated and the skeleton is drawn.

```
Video/USB/CSI → PeopleNet → NvDCF Tracker → BodyPose3DNet
     → skeleton rendering → preview
```

The key point estimation is affected by the camera viewpoint, the occlusion, the detection boxes and the training domain; it is not a medical-grade measurement.

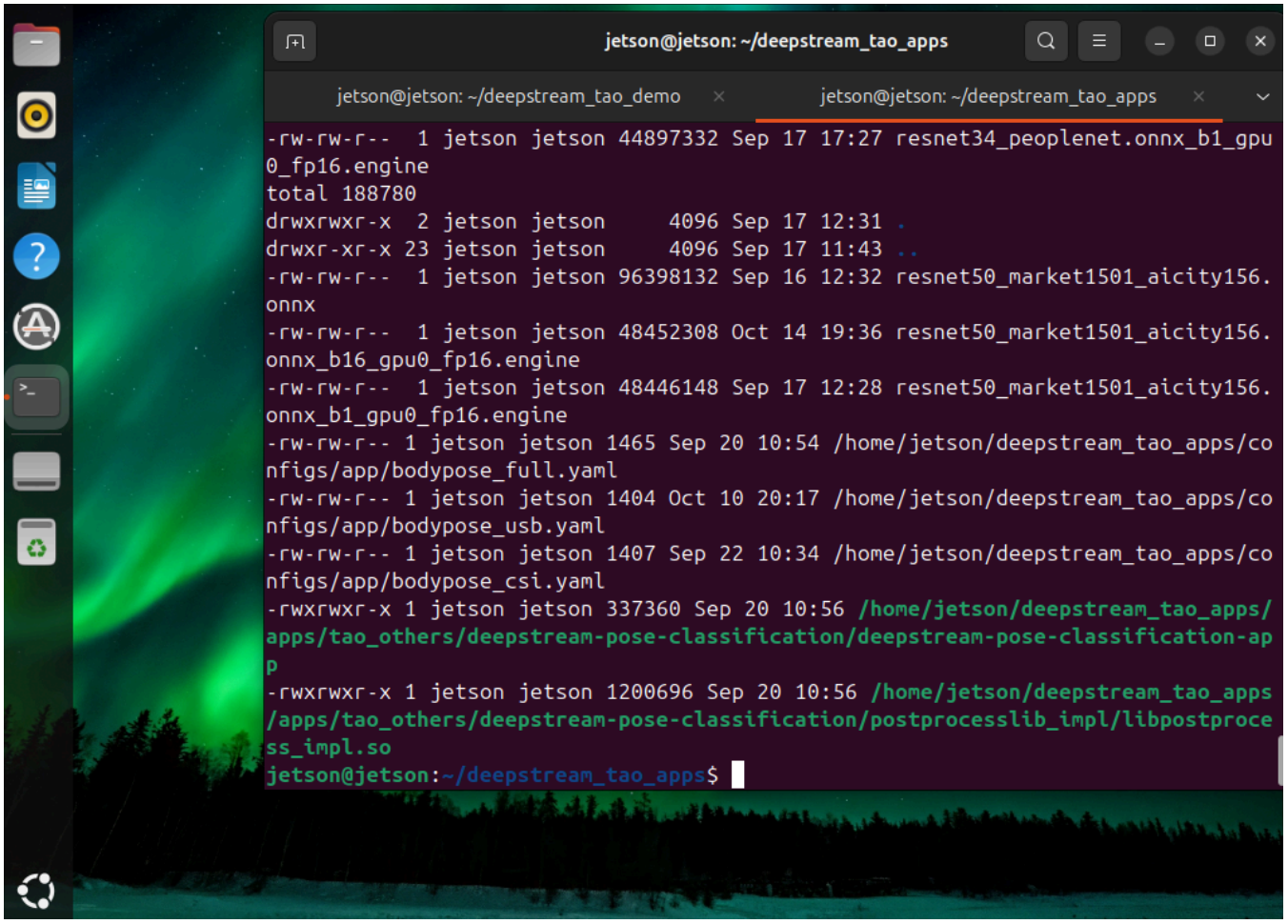
2. Checks before running

```
cd ~/deepstream_tao_apps

# Confirm the model directories
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/bodypose3dnet_vdeployable_accuracy_onnx_1.0/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/peoplenet_vdeployable_quantized_onnx_v2.6.3/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/reidentificationnet_vdeployable_v1.2/

# Confirm the YAML configurations
ls -la ~/deepstream_tao_apps/configs/app/bodypose_only.yaml
ls -la ~/deepstream_tao_apps/configs/app/bodypose_usb_noclass.yaml
ls -la ~/deepstream_tao_apps/configs/app/bodypose_csi_noclass.yaml

# Confirm the compiled artifacts of the dedicated app
ls -la ~/deepstream_tao_apps/apps/tao_others/deepstream-pose-classification/deepstream-pose-
classification-app
ls -la ~/deepstream_tao_apps/apps/tao_others/deepstream-pose-
classification/postprocesslib_impl/libpostprocess_impl.so
```

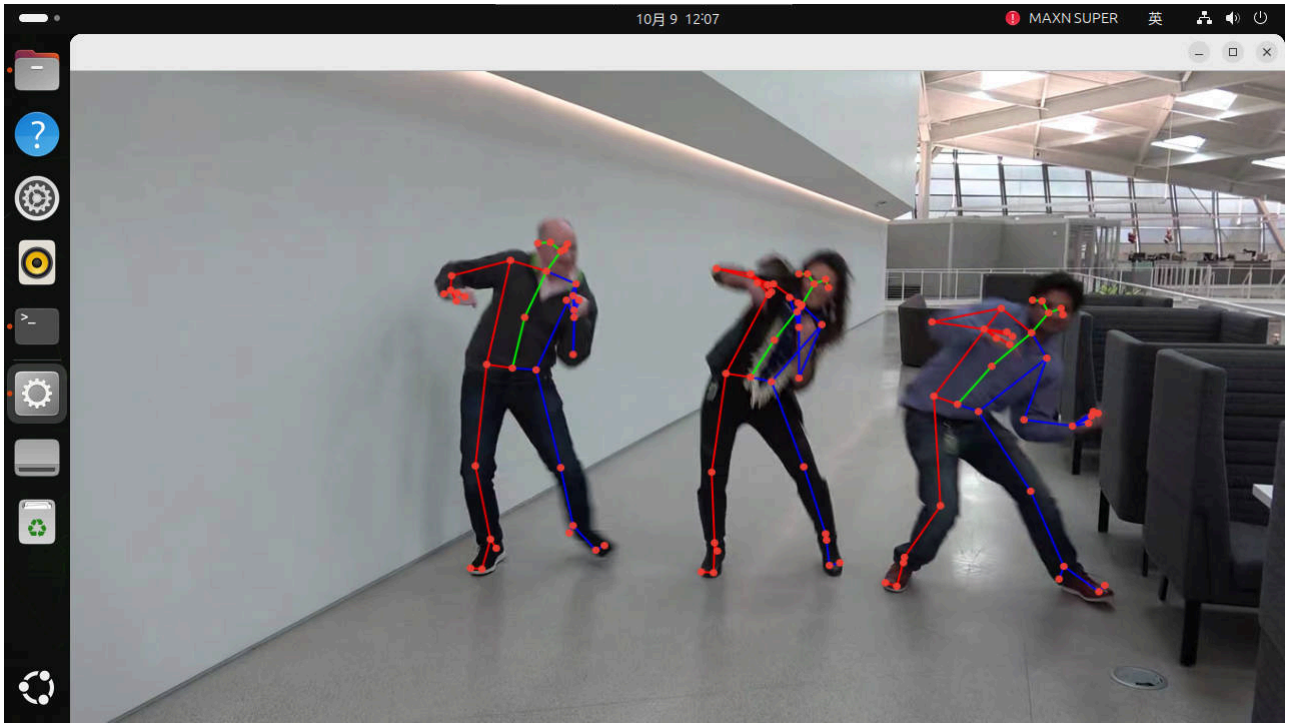


Run it in the graphical session of the Jetson desktop/VNC and confirm that the input and the output directories are available.

3. Starting the detection

Video mode:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_bodypose3d.sh
```



The entry point uses `~/deepstream_tao_apps/configs/app/bodypose_only.yaml`, and the default input is the `sample_720p.h264` that comes with DeepStream. This mode turns off the action classification, the detection boxes, the text and the frame number, and it only keeps the skeleton.

Or pass the parameter directly to replace the video:

```
~/deepstream_tao_demo/run_bodypose3d.sh /home/jetson/Videos/your_video.mp4
```

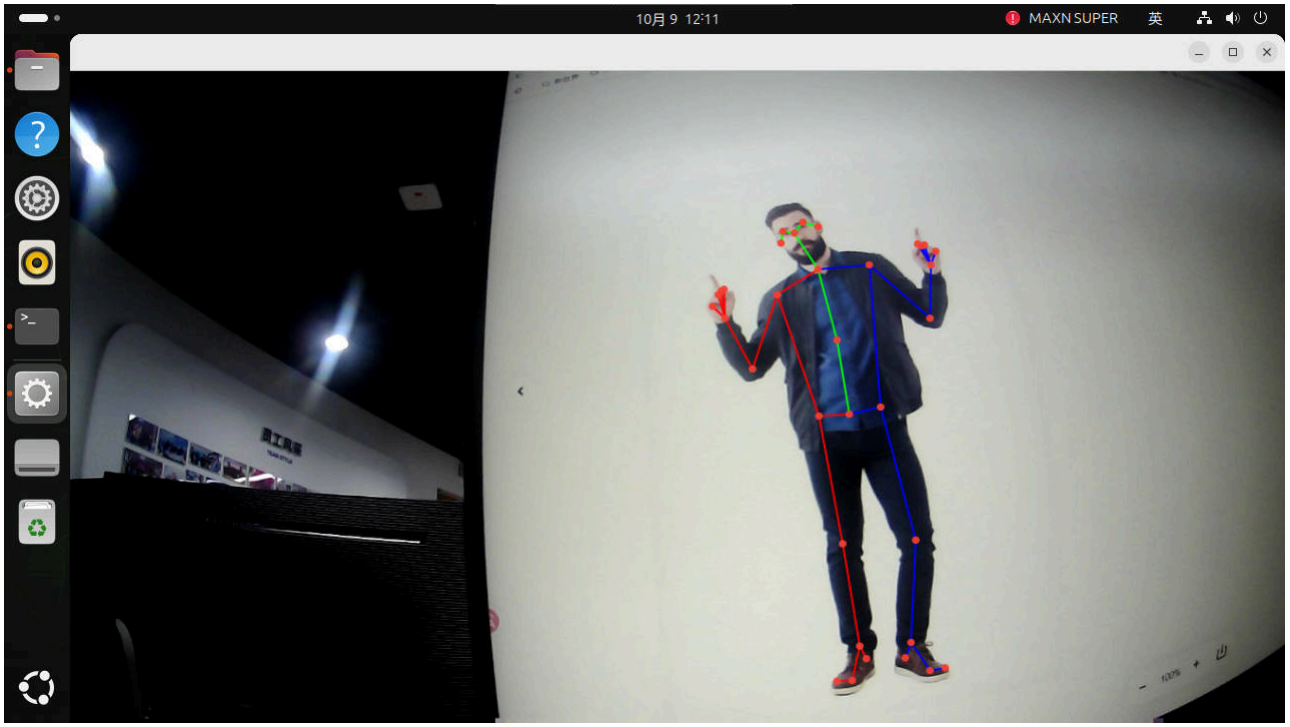
To replace the test video: preferably pass the absolute path directly to the script, or modify the URI in the `source-list.list` of `bodypose_only.yaml`; the new material is best if it contains a clear and complete human body.

4. Real-time detection with a USB camera

After confirming that the USB camera device node exists, run:

```
ls -l /dev/video*
v4l2-ctl -d /dev/video0 --list-formats-ext

cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_bodypose3d.sh usb
```



The script detects the USB capture node and the supported resolution/frame rate automatically and generates a temporary YAML. The USB mode is a real-time preview and it does not produce an offline MP4.

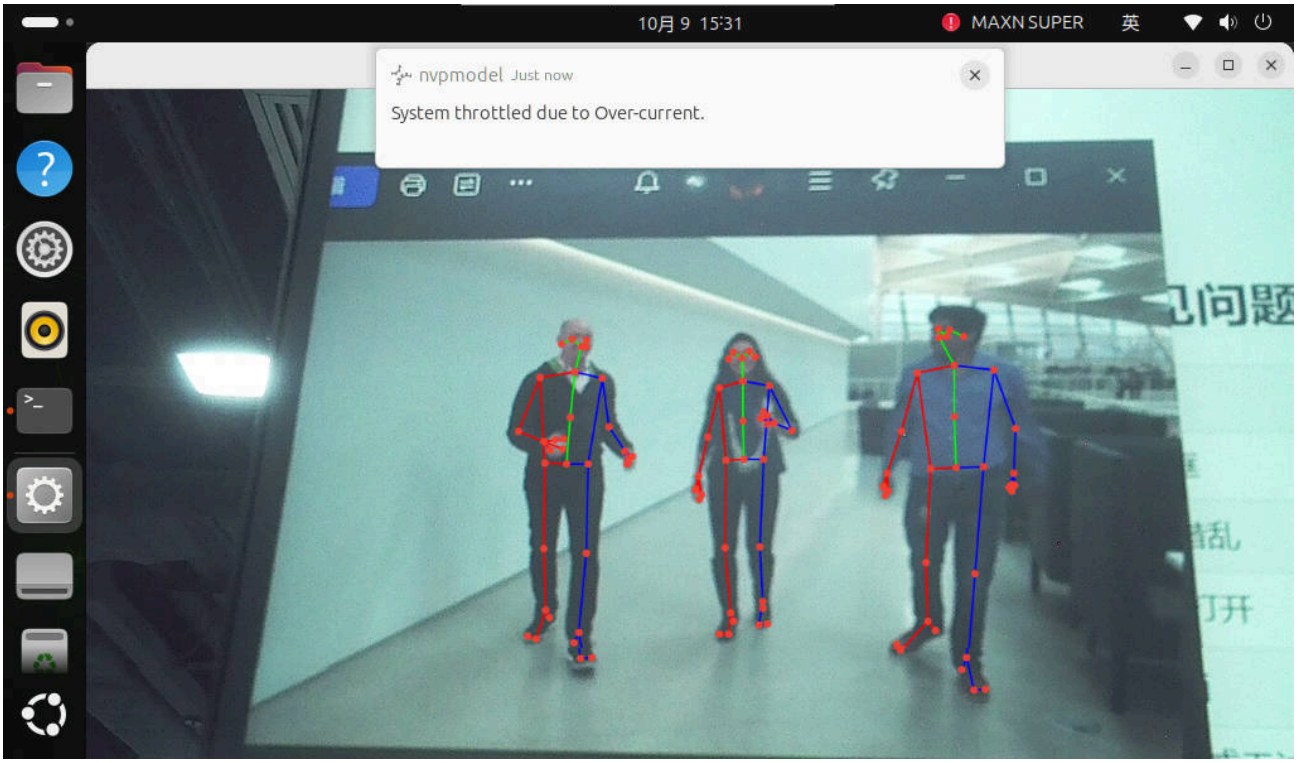
To stop it, press `Ctrl+C` once and wait for the application and the camera to exit.

5. Real-time detection with a CSI camera

The CSI mode supports a real-time preview. The script uses Argus directly and it requests `sensor-mode=3` and the automatic white balance `wbmode=1` by default; when the plugin does not support `sensor-mode`, the `1920x1080@30` caps selection mode is used.

Start:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_bodypose3d.sh csi
```



USB/CSI is a continuous stream and it does not naturally play to the end. To stop it, press `Ctrl+C` once in the terminal that started it and wait for the application and the camera to exit before you start another camera task.

6. Common issues

Symptom	Check and handling
No skeleton	Check PeopleNet, the Tracker, the BodyPose3DNet and the model loading log in turn; the model comes from the DeepStream `samples/models/tao_pretrained/bodypose3dnet_vdeployable_accuracy_onnx_1.0/`.
The skeleton lines jitter	The BodyPose3DNet has limited robustness for distant scenes/occlusion, so the jitter is normal.
There is a skeleton but no person box/action label	This is the expected effect of the pure 3D mode; if you need the detection boxes and the action labels, run `run_poseclass.sh`.
No picture with USB	Check the node, the format, the occupancy, the permissions and the graphical session; the resolution has to match the capsfilter in the YAML.
No picture with CSI	Check the compatible module, the Argus service, the `sensor-id` and the camera occupancy; you can run the `nvargus-daemon` status command to troubleshoot.
`No cameras available`	Check the CSI ribbon cable, the `sensor-id=0` and `nvargus-daemon`.
The `[PREPROC]` / `[POSE-CLASS]` log	This is the normal informational output of the pose post-processing, not an error.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1 and it does not affect the inference and the display.

7. References

- NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html